

Test Document: Vector Databases

A Technical Overview for Wiki Integration

Abstract

Vector databases enable semantic search by storing numerical representations of documents called embed

Introduction

Traditional search relies on keyword matching, which fails when users express concepts differently than aut

Vector databases solve this by converting text into high-dimensional vectors using embedding models. Simi

Embedding Models

Popular embedding models include:

1. all-MiniLM-L6-v2 (384 dimensions, 80MB)
A lightweight model by Sentence-Transformers, ideal for CPU deployment.
Produces 384-dimensional vectors with good semantic accuracy.
2. text-embedding-3-small (1536 dimensions)
OpenAI's embedding model, API-based, higher accuracy but requires network access and has per-token costs.
3. bge-large-en-v1.5 (1024 dimensions)
BAAI's general-purpose embedding model, strong performance on MTEB benchmarks, requires GPU for reasonable throughput.

Indexing Strategies

Vector databases use approximate nearest neighbor (ANN) algorithms to efficiently search through millions

- HNSW (Hierarchical Navigable Small World): Graph-based, provides excellent recall with logarithmic search complexity.
- IVF (Inverted File Index): Clusters vectors into partitions, searches only the most relevant clusters.
- PQ (Product Quantization): Compresses vectors for memory efficiency at the cost of some accuracy.

Practical Applications

1. Semantic Document Search

Replace keyword-based search with meaning-based retrieval. Users can ask natural language questions and find relevant documents.

2. Retrieval-Augmented Generation (RAG)

Augment LLM responses with relevant context retrieved from a vector database, reducing hallucinations and grounding answers in facts.

3. Recommendation Systems

Find similar items based on their vector representations. Used by e-commerce platforms, streaming services, and social networks.

4. Anomaly Detection

Detect unusual patterns by comparing new vectors against historical clusters in vector space.

Conclusion

Vector databases represent a fundamental shift in how we store and retrieve information. By operating on s